Hyperspectral Image Stripe Removal Network With Cross-Frequency Feature Interaction

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***Abstract*— Remote sensing images, especially hyperspectral images (HSIs), are extremely vulnerable to random noise and stripe noise. As a key aspect of HSI data quality improvement, stripe noise removal has always been a pervasive issue in remote sensing image processing. Convolutional neural networks have been applied for HSI data destriping. However, the existing methods lose the stripe-free component of the original image to a certain extent. These models also ignore the global spatial context of images and the correlation between spatial information and spectral information. Therefore, we propose a novel destriping convolutional network to overcome the problems with the existing methods. Octave convolution is used to extract cross-frequency features, and separate and compress the low-frequency information of the images, while dilation convolution (Dila-Conv) is used to reduce the amount of required calculation and also preserve the key image information. In addition, Dila-Conv can expand the receptive field to obtain multiscale features. Finally, a cross-channel enhanced spatial–spectral feature fusion module is used to acquire and integrate spatial context information and interchannel dependencies on a global scale as auxiliary information so that the network model can learn and pay attention to key feature information, specifically, "what to look for" and "where to look at," which can facilitate the distinction between stripe and stripe-free components. Experimental results obtained using multiple datasets demonstrated that the proposed method can outperform the existing comparable methods and can produce satisfactory results in terms of visual effects and quantitative evaluation.**

***Index Terms*— Attention mechanism, convolutional neural networks (CNNs), destriping, hyperspectral images (HSIs).**

I. Introduction

HYPERSPECTRAL image (HSI) data are widely used in many fields, including urban planning, ground object classification, and land detail inversion, due to their advantages such as map unification and high spectral resolution. However, sensor malfunctions, atmospheric interference, ground surface, and other degradation factors are prone to causing degraded or missing radiation information in HSI data, resulting in problems such as stripe noise and random noise, which reduce the HSI's data utilization rate. Therefore, image destriping is typically adopted as a preprocessing step to reduce the noise in HSIs and improve their quality before HSI analysis and interpretation [1], [2].

Stripe noise is a common type of noise in hyperspectral data and is usually caused by factors such as sensor instability, inconsistent sensor unit response, and charge-coupled device (CCD) dark current interference [3]. From previous research [4], according to its characteristics, stripe noise can be systematically divided into six categories: regular periodic stripes, periodic stripes with discontinuity, nonperiodic stripes with random width, nonperiodic stripes with random length, mild stripes with mild random noise, and severe stripes with severe random noise. The presence of stripe noise conceals the true image information and limits the application of HSI data.

Numerous studies on stripe removal have been conducted. The present mainstream methods of destriping HSI data have been derived from gray-scale or multichannel image denoising methods. Most perform band-by-band processing to achieve stripe removal. For example, the nonlocal means filtering [5], block matching and 3-D filtering [6], the Fourier transform [7], and wavelet decomposition [8] methods use filtering to separate clean images and stripes. Furthermore, methods involving statistical histogram matching utilize the statistical characteristics of image pixels to perform destriping [9], and other methods remove stripes based on low rank and sparse unmixing [10]. In recent years, with the development of deep learning methods, convolutional neural networks (CNNs) have provided new approaches for improving the quality of HSI data with their powerful feature learning and representation capabilities. A method of HSI data destriping utilizing CNNs has gradually emerged [11]. However, multiple problems remain.

1) HSI has numerous bands, their stripe noise is more complex, and the correlation between bands is strong, which has a certain effect on the destriping performance; therefore, how to employ spatial information and spectral information fully to improve the effect of stripe removal is a key issue.
2) The existing CNN stripe removal methods focus more on stripe removal and largely ignore the destriped image distortion caused by spectral information loss; therefore, in-depth research and investigation on how to balance

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the destriping effect and retain the clean components of the image are of great significance to HSI destriping.

3) The spatial location distribution is an important sign of stripe noise. There is a spatial dependence among pixels, such as long-distance dependence, which plays a key role in the distinction between noisy and clean pixels. The use of spatial dependence to assist HSI stripe removal is also worthy of investigation.

In response to the problems mentioned above, we propose En-OCTNet, a neural network based on octave convolution (Oct-Conv) [12] for HSI data destriping that uses the developed version of Oct-Conv. In Oct-Conv, the feature extraction is divided into two branches to extract high- and low-frequency features and exchange the information between different frequencies, which is also more compatible with the destripe task. A cross-channel enhanced spatial–spectral information fusion module is applied to integrate the channels and global context information in order to improve destriping performance. The contributions of our work can be summarized as follows.

1) To balance the destriping performance and retain the stripe-free component, the feature extraction is divided into two branches to extract high- and low-frequency features and frequency communication by Oct-Conv. Furthermore, dilation convolution (Dila-Conv) is used instead of the pooling layer for better preservation of the stripe-free component and to obtain a larger receptive field. The whole En-OCTNet takes advantage of cross-frequency feature extraction and interaction to better retain the HSI spectral information for destriped images.
2) A spatial–spectral attention mechanism is used for excavating the intrinsic feature between the HSI's stripe component and stripe-free component. Moreover, a cross-channel enhanced information exchange strategy is applied to better exploit the HSI data channel correlation. This approach strengthens the stripe removal performance of this method in different scenarios.
3) Two experiments, the first conducted on a public HSI dataset (simulation) and the second on real data, have demonstrated the feasibility and advantages of the proposed method.

The remainder of this article is organized as follows. Section II describes the details of the HSI stripe model and removal methods. Section III presents the architecture and details of the proposed model, while Section IV discusses the simulated and real data experimental results. Finally, Section V summarizes the conclusions of this study.

II. Related Work

Complimentary to the brief summary of existing HSI data stripe removal methods in the previous section, this section systematically summarizes the status quo of those methods. The existing methods can be roughly classified into two groups: traditional destriping methods and CNN-based destriping methods.

A. Traditional Destriping Methods

We consider the destriping methods before the advent of CNNs to be "traditional" methods. Traditional methods have a wide range of applications in the HSI data destriping field due to their advantages of being fast, lightweight, and nondata-driven. Traditional methods have two common characteristics: being driven by prior knowledge and using handcrafted features extraction. Among them, Horn and Woodham [13] proposed a statistics-based method that utilized histogram matching to eliminate stripes in Landsat images. Corsini *et al.* [14] designed a statistics-based destriping method for MOS-B data, which assumes that an image is a quasi-homogeneous target and estimates equalization curves for stripe removal by utilizing carefully selected homogeneous targets. However, the histogram matching method is sensitive to violations. Therefore, Gadallah *et al.* [15] claimed that the radiation distribution of ground objects on different sensors is consistent, meaning that each sensor has the same mean and standard deviation. These two typical values, which can be obtained by applying an alternative algorithm, can achieve destriping.

Filter-based methods are one of the most representative methods in traditional methods, which use stripe components and stripe-free components to show different frequency characteristics in the frequency domain and design filter operators to achieve the purpose of destriping [16]. For example, Rakwatin *et al.* [17] designed a destriping method that combines histogram matching with an iterated weighted least-squares facet filter for MODIS data. Li *et al.* [18] reformulated the 1-D Vondrak filter to a 2-D filter and simultaneously used conjugate gradients to solve the least-squares optimization problem, which could smooth out the noise while preserving the original image information. Wavelet transformation is commonly employed in digital image processing [19] and is another powerful method for removing noise in HSIs.

The sparseness and low rank of stripe-free images have been widely used as prior knowledge in traditional destriping methods. For example, Liu *et al.* [20] proposed a method based on a 3-D wavelet transform and low-rank prior to remove stripe noise from HSIs acquired by pushbroom satellites. Xue *et al.* [21] proposed a joint spatial and spectral low-rank regularized HSI data denoising method based on the different characteristics of noisy and noise-free signals in a low-dimensional structure. Furthermore, using the characteristics of stripe-free HSIs image low-rank and sparse, Xie *et al.* [22] proposed a low-rank and sparse matrix decomposition method based on nonconvex regular optimization to remove hyperspectral data stripes. In addition, combining low rank and sparseness with total variation for HSI denoising is a common method [23]–[26]. Wang *et al.* [27] proposed a multidirectional low-rank modeling and spatial and spectral total variation (MLR-SSTV) method to overcome the loss of structure using a stripe-free HSI image total variation that was smaller than the stripe-containing image information and spatial and spectral relationship.

As mentioned before, typical traditional destriping methods use a well-designed feature extraction operator based on prior knowledge. Hence, the feature extraction operation is limited by prior knowledge, and the extracted features are also accordingly limited; moreover, deep image features cannot be mined. In addition to such limitations, it is necessary to design

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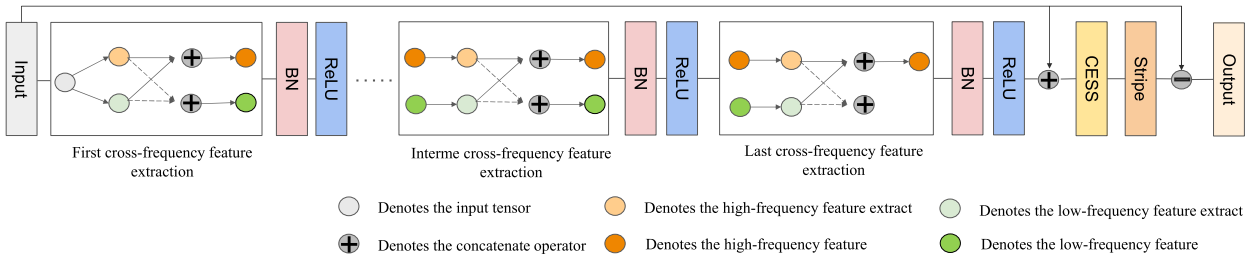


Fig. 1. Proposed En-OCTNet model. CESS denotes the cross-channel enhanced spectral–spatial feature fusion.

a destriping method for different stripe types, as destriping in traditional methods is not sufficiently versatile and cannot handle complex stripes well.

B. CNN-Based Destriping Methods

Convolutional neural-network-based methods have been successfully applied in computer vision and have been introduced in remote sensing image processing. Such methods have also been used in the applications of remote sensing image land cover classification [28], semantic segmentation [29], object detection [30], and more. These approaches provide both new and long-established techniques for solving classic problems and have achieved reasonable success in applications. In recent years, CNN-based methods have gradually emerged in denoising and destriping. Due to their powerful feature learning abilities, convolutional networks have great potential in learning to distinguish between stripe- and stripe-free component features.

At present, most HSI data destriping networks are one- or two-stream networks, and the backbone used generally consists of a residual network and Unet. For example, Chang *et al.* [31] proposed a two-stream network combined with wavelets for single HSI destriping based on the Unet structure but only considered the spatial features of HSIs and ignored the channelwise contribution to stripe removal. Liu and Lee [32] proposed a two-stream noise removal network for replacing the superposition of multiple convolution modules to reduce the number of parameters and expand the receptive field through Dila-Conv to obtain multiscale image information.

In addition, the one-stream structure network is also applied to HSI destriping. For example, Yuan *et al.* [33] extracted the multiscale features of the noisy band and its adjunct K-bands and improved the restoration precision by employing the information of the neighboring bands. Maffei *et al.* [34] trained a noise level map as input data based on the methods of Yuan *et al.* [33] and FFDNet [35], which can deal with complex noise situations and balance noise removal and detail repair. Furthermore, Chang *et al.* [36] proposed an HSI-DeNet utilizing residual learning based on DnCNN [37], which can remove many types of stripe noise. There are also methods that include adding image gradient information to a network, such as the networks proposed by Zhong *et al.* [38] and Zhang *et al.* [39], which takes the spectral data and image direction gradient information as the network input to remove stripe noise.

In particular, the attention mechanism plays a critical role in computer vision tasks. In recent years, important breakthroughs have been made in the field of image processing, and the benefits of improving model performance have been proven. The attention mechanism also has some applications in the field of HSI destriping or denoising. For example, Shi *et al.* [40] introduced a dual-attention mechanism, which uses spectral information and spatial information to enhance the denoising of HSI data. Ma *et al.* [41] applied an enhanced nonlocal structure network and a channel attention module to enhance extracted multiscale features for HSI denoising and Zhao *et al.* [42] proposed an attention-based deep residual network for HSI denoising. In general, the main purpose of the applied attention-based mechanism is to utilize the spectral–spatial information and extract nonlocal information for better denoising performance.

III. Methodology

To better balance the removal stripes of HSIs, retain the stripe-free component, and excavate the distinguishing feature between HSIs' stripe- and stripe-free components, En-OCTNet is proposed. As illustrated in Fig. 1, En-OCTNet can be divided into two modules: cross-frequency interactive feature extraction and enhanced cross-channel spectral–spatial (CESS) feature fusion. Our method consists of 17 cross-frequency interactive feature extraction layers and appends cross-channel enhanced spectral–spatial feature fusion. Full details are provided in the following.

A. HSI Data Stripe Noise Modeling

The HSI can be regarded as a 3-D tensor Y of size $R \times C \times B$, and the degradation model of the HSI can be described as

$$Y = X + N \tag{1}$$

where X denotes the clean data; N denotes noise in HSI, which phrases stripe noise in this article; and R, C, and B denote the height, width, and depth of the HSI data, respectively. The final purpose of HSI data destriping is to obtain stripe-free data X from degraded data Y.

B. Cross-Frequency Interactive Feature Extraction

Generally, in the frequency domain, the image information is expressed by signals of different frequencies; high frequencies correspond to detailed structure or noise information, whereas low frequencies correspond to the main image information. Thus, Chen *et al.* [12] had proposed Oct-Conv, which has the two-branch structure to both capture the

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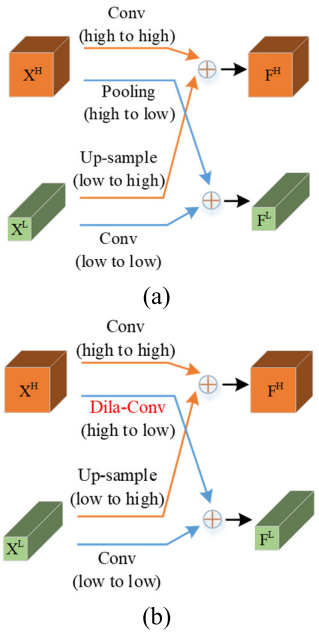


Fig. 2. (a) Oct-Conv and (b) developed Oct-Conv proposed in this article, which uses Dila-Conv. Superscripts H and L denote the high and low frequencies, and X^H, X^L, F^H, and F^L denote the high- and low-frequency input and output features, respectively.

different frequencies that represent the detailed structure or noise information and also main image information, as shown in Fig. 2(a).

In detail, let X be the input data; $\{F^H, F^L\}$ be the output data; superscripts H and L denote the high and low frequencies, and X^H, X^L, F^H, and F^L denote the high- and low-frequency input and output features, respectively. The output tensors $F^H = F^{H\to H} + F^{L\to H}$ and $F^L = F^{L\to L} + F^{H\to L}$, due to the low-frequency capture branch, reduce the spatial resolution to preserve the low-frequency feature, which can both reduce the network parameters and channelwise redundancy; meanwhile, the receptive field is enlarged. Particularly, to aggregate information of different frequencies, a mechanism for exchanging information between frequencies is established to facilitate information communication between the two octaves, where $F^{H\to H}$ and $F^{L\to L}$ denote the intrafrequency updates, and $F^{H\to L}$ and $F^{L\to H}$ denote the interfrequency communications. The relevant formulas in Oct-Conv are given as follows:

$$\begin{aligned} F^H &= F^{H\to H} + F^{L\to H} \\ &= \sum \left(W^{H\to H}\right)^T X^H + \text{upsample}\left(\sum \left(W^{L\to H}\right)^T X^L\right) \end{aligned} \tag{2}$$

$$\begin{aligned} F^L &= F^{L\to L} + F^{H\to L} \\ &= \sum \left(W^{L\to L}\right)^T X^L + \text{pool}\left(\sum \left(W^{H\to L}\right)^T X^H\right) \end{aligned} \tag{3}$$

where T is the transposition operation and W is the weight matrix. To calculate the relevant variables in Oct-Conv, W is decomposed into two components to extract cross-frequency features, that is, $W = [W^H, W^L]$, where W^H and W^L can be further decomposed into $W^H = [W^{H\to H}, W^{L\to H}]$ and $W^L = [W^{L\to L}, W^{H\to L}]$. Feature extraction is performed separately on X^H and X^L, while $F^{L\to H}$ is obtained by upsampling, and $F^{H\to L}$ is obtained by pooling.

The main functions of the pooling layer are to reduce dimensionality, expand the receptive field, prevent overfitting and translation invariance, and so on. However, for pixel-level image restoration tasks, an obvious defect in performing pooling for downsampling is that the compression of the feature space leads to the loss of image information, such that all information cannot be completely restored during upsampling, resulting in image distortion after destriping. Inspired by previous research [32] and [43], it is, herein, proposed that Dila-Conv be used to replace the downsampling-based pooling layer in the original Oct-Conv, such that a large receptive field can be produced while retaining the original information, as shown in Fig. 2(b). The parenthetical references represent the cross-frequency feature extraction operations; for example, "high to low" denotes interfrequency communication between the high- and low-frequency feature maps. Thus, the proposed network can retain the original image information to the greatest extent to reduce the loss of detail.

C. Cross-Channel Enhanced Spectral–Spatial Feature Fusion

Stripes have global characteristics with obvious spatial distributions and shape characteristics. Most stripes have certain lengths and widths, and are randomly distributed in space. Employing contextual information and long-distance dependency among pixels is a key method for destriping. Dila-Conv can capture the context information between adjacent pixels to a certain extent by expanding the receptive field, but the long-distance dependence on the global scale must be stacked through multiple convolutional layers of different scales. However, this requirement leads to the high computational complexity of the network, making optimization difficult [44]. Therefore, to better capture the global context information and channelwise dependency, the CESS component is established. This component assists in understanding the visual scene, strengthens the cross-channel information exchange, and is able to learn the discriminative features and instruct the model "what to look for and where to look at."

As shown in Fig. 3, the CESS contains two subparts: global context attention modeling and feature fusion. The attention modeling component is used to model the spatial global context, while the feature fusion component is used to aggregate the spatial and spectral features, which includes feature transform and the cross-channel enhanced module. The cross-channel enhanced module is used to further explore the dependence between channels and achieve cross-channel information exchange of the feature map.

In detail, the CESS includes three steps: 1) global context modeling; 2) using feature transformer for feature squeeze in the channel domain; and 3) recalibration of the importance of each channel via the cross-channel enhanced module to obtain the interchannel dependence and broadcast elementwise addition for feature fusion. The CESS, which adopts the ideas of nonlocal networks [45] and SENet [46], can model long-distance dependencies, such as nonlocal networks, and are lightweight and simple. Unlike the original nonlocal network, the attention modeling component first obtains the global

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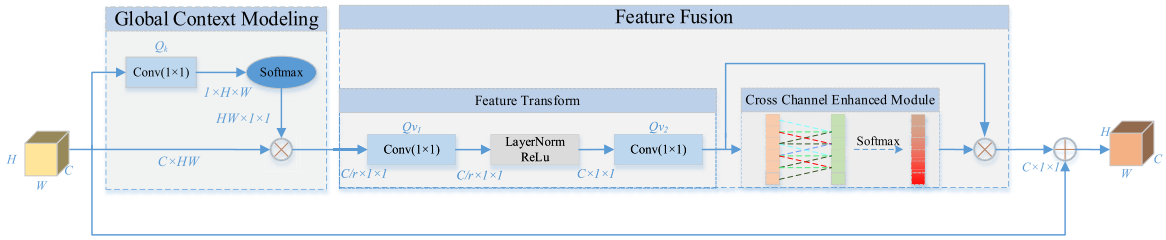


Fig. 3. Attention modeling framework, where the channel number, height, and width of the feature map are denoted by C, H, and W, respectively. $\otimes$ and $\oplus$ denote matrix multiplication and broadcast elementwise addition, respectively, and r is the bottleneck ratio, which was set to 16.

attention weight through a 1×1 convolution and softmax operation, and acquires the global attention map through matrix multiplication with the input feature. All position pixels share this global attention map to reduce the amount of calculation. Then, a bottleneck transform is performed to capture the interchannel dependencies. Because the bottleneck transform increases the optimization difficulty, layer normalization is performed before ReLU to reduce the optimization difficulty and improve the generalization ability. In order to more accurately obtain the interdependence between channels, the proposed module applies a cross-channel enhanced module, which utilizes 1-D convolution with a kernel size of 3 to establish the cross-channel information interaction and enhance the spectral–spatial feature fusion. Finally, addition is employed to aggregate the global context features to the features of each position.

Suppose that X is the input feature map; Z is the output feature attention map; i, j, and m represent the pixel position index; N is the total number of pixels; and Q_k, Q_{v1}, and Q_{v2} represent linear transformation matrices, which can be formulated as

$$Z_i = X_i + Q_{\mathrm{CE}}\left(Q_{v2}\mathrm{ReLu}\left(\mathrm{LN}\left(Q_{v1}\sum_{j=1}^{N}\frac{e^{Q_k X_j}}{\sum_{m=1}^{N} e^{Q_k X_m}}X_j\right)\right)\right) \tag{4}$$

where $(e^{Q_k X_j}/\sum_{m=1}^{N} e^{Q_k X_m})$ denotes the global attention weight, $\sum_{j=1}^{N}(e^{Q_k X_j}/\sum_{m=1}^{N} e^{Q_k X_m})X_j$ denotes the global context modeling, $Q_{v2}\mathrm{ReLu}(\mathrm{LN}(Q_{v1}(e^{Q_k X_j}/\sum_{m=1}^{N} e^{Q_k X_m})X_j))$ denotes the bottleneck transform used to channel model, and Q_{CE} denotes the cross-channel enhanced module to capture cross-channelwise dependencies. As shown in (4), the global context modeling is not constrained by pixel location; rather, the global context is directly modeled as the weighted average of all location features, and the addition operation is then performed with the features of each location and the global context features as the final output feature attention map.

IV. Experimental Results and Discussion

To validate the effectiveness of the proposed method, in this section, the experimental process is described in detail, including the following: data augmentation, basic training parameter settings, evaluation index with comparison methods, simulation experimental design, experimental datasets, and results obtained with both simulated and realistic stripe removal.

A. Data Augmentation and Parameters' Settings

We referred to the processing methods in [36] and [38] to process the dataset. First, we normalized the data to the range (0, 1) and then cropped the training data into subcubes of size $40 \times 40 \times 10$. Considering that insufficient training data would cause model overfitting, we performed data augmentation, including multiangle rotation (angles of 0°, 90°, 180°, and 270°), horizontal flipping, and vertical flipping. The data were shuffled before each epoch was started. The size of all convolution kernels was set to 3×3, with a stride of 2 and a dilation rate of 2, and we padded each layer with the same size as the original image. We trained the model for 300 epochs with a batch size of 24 and a fixed learning rate of 0.001. We used the Adam [47] as the optimizer with the parameters β_1, β_2, and ε, which were set to 0.9, 0.999, and 1e–8, respectively. Here, we used the mean-square error loss (MSELoss) as the loss function.

B. Comparison Methods and Evaluation Index

We selected a statistics-based method (statistical linear destriping (SLD) [9]), a filter-based method (wavelet-Fourier adaptive filter (WFAF) [49]), a total variation method (UTV [50]), a low-rank method (low-rank tensor decomposition (LRTD) [51]), and three deep learning-based methods (HSI-DeNet [36], SGIDN [38], and MsDUC [52]). These methods were implemented using open-source codes, and the parameters were set to the recommended values. For fairness, we used the training data of this study to retrain the HSI-DeNet, SGIDN, and MsDUC methods and employed the trained models for comparison.

To evaluate the test results, we utilized the following quality evaluation indicators commonly used in the field.

1) Simulation Experiment Evaluation Index: We employed two indicators that are widely used in HSI data quality evaluation for simulation experiment results evaluation: the peak signal-to-noise ratio (PSNR), the structural similarity index measurement (SSIM) [53], and the spectral angle map (SAM) [54]. The calculation formulas are given as follows:

$$\mathrm{PSNR} = 20\log_{10}\left(\frac{\mathrm{MAX}_I}{\sqrt{\mathrm{MSE}}}\right) \tag{5}$$

where MAX_I is the maximum pixel value to be evaluated and MSE is the mean square error of the reference image. The larger the PSNR, the better the destriping effect; and

$$\mathrm{SSIM} = \frac{(2\mu_X\mu_Y + c_1)(2\sigma_{XY} + c_2)}{(\mu_X^2 + \mu_Y^2 + c_1)(\sigma_X^2 + \sigma_Y^2 + c_2)} \tag{6}$$

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where X and Y are the image to be evaluated and the reference image, respectively; μ_X and μ_Y are the pixel mean values of images X and Y, respectively; σ_X and σ_Y are the variances of images X and Y, respectively; and σ_{XY} is the covariance of images X and Y. c_1 and c_2 are the constant values of 6.5025 and 58.5225, respectively. The larger the SSIM, the more similar the two images, and hence, the better the destriping performance. To evaluate the destriping performance overall, we evaluated the quality of all bands utilizing the mean PSNR (MPSNR), mean SSIM (MSSIM), and mean SAM (MSAM).

2) Real Data Test Evaluation Index: We selected the inverse coefficient of variation (ICV) [55] and mean relative deviation (MRD) [56] as experimental evaluation indicators for the real data. The higher the ICV, the better the stripe removal performance, whereas the opposite was true for the MRD. We selected a homogeneous area of a certain size to calculate the ICV and MRD.

C. Simulation Experiment Results

We used the Washington DC Mall HSI dataset for the simulation experiments. The dataset, obtained by the Hyperspectral Digital Imagery Collection Experiment airborne sensor, with a size of $1280 \times 303 \times 191$, was provided by the Purdue University Research Repository. We divided these data into two parts, for which one part with a size of $1080 \times 303 \times 191$ was used for training, and the other part with a size of $200 \times 200 \times 191$ was used for testing.

We designed and simulated five types of stripes for simulation experiments, denoted as Cases 1–5. The details are given as follows.

Case 1: Add a fixed period of stripe noise with a standard deviation of 20, 30, or 50. The stripe noise intensities of different bands are the same.

Case 2: Add stripe noise with a standard deviation from 0 to 50. The stripe noise intensity is uniform in each band, but the stripe noise intensities of different bands are different.

Case 3: Add nonperiodic, random length stripe noise with a standard deviation of 0–20. The stripe noise intensities of different bands are the same.

Case 4: Add nonperiodic, random width stripe noise with a standard deviation of 0–20. The stripe noise intensities of different bands are the same.

Case 5: Stripe noise is mixed with the Gaussian noise with a standard deviation from 0 to 20. The stripe noise intensities of different bands are different.

Here, the simulated stripe noise for each band is generated by a set of numbers that obey Gaussian distribution with mean μ and standard deviation σ, and the size is $1 \times R$ and then extend the $1 \times R$ to $C \times R$. Here, R and C denote the height and width of the HSI data.

Table I lists the evaluation results of the simulation experiment. The evaluation indicators listed in Table I demonstrate that the proposed destriping method, En-OCTNet, achieved the best results, with the ability to remove different stripe levels and types of stripes well. In particular, for stripes of random length, width, and mixed noise, EN-OCTNet exhibits outstanding results. To show the destriping performances of the different methods more intuitively, we selected a few representative results to depict in pseudocolor image either with bands 57, 27, and 17 or in gray-scale images. For stripes with different fixed period stripe levels, as shown in Fig. 4 for Case 1 with $\sigma = 50$, the SLD method does not achieve good stripe removal, and there is a certain degree of stripe residue; as the stripe level increases, the stripe residue becomes more obvious. The UTV method exhibits an obvious oversmoothing phenomenon. The reconstructed images are distorted after stripe removal using the WFAF method and cannot retain the spectral information of the original image. The destriping effect of LRTD is more obvious, and the spectral information can also be retained to a greater extent. However, as the stripe level increases, there will be remaining stripes that cannot be completely removed. Most stripes could be removed after processing by MsDUC and SGIDN; hence, these methods are better than the other methods in terms of visual perception, and the restored images are closer to the original images. However, as the stripe intensity increases, the remaining stripes in the repaired images become more obvious. For HSI-DeNet, the destriping effect is better visually, but, for the repaired image, the recovery effect of the spectral information is not good, particularly the spectral information of the buildings in the picture, which is not well recovered. Overall, the destriping results of En-OCTNet are best. The proposed method achieves decent destriping results and simultaneously retains the original spectral information, yielding results that are closest to the original image among those obtained using the other investigated approaches.

In Case 2, as shown in Fig. 5, obvious residual stripes remain when the three traditional methods are used. In particular, the UTV method exhibits stripe residuals and an oversmoothing phenomenon. After MsDUC processing, the effect is not satisfactory, and many stripes remain. After processing by HSI-DeNet, a better visual effect can be obtained, but the recovery effect of the spectral information is slightly inferior to the proposed method, and there is a noticeable difference from the original spectral information, as shown in the enlarged area in the figure. After SGIDN processing, the stripes are not completely removed. Regarding the image restoration, the spectral information of some feature details is still different from that in the original image features. Figs. 6–8 present the results for Cases 3–5, respectively, which are the three more complicated stripe cases. In these cases, the proposed method can still achieve satisfactory destriping and can be applied to stripes with random length or width and random noise. In contrast, when the stripe length is random, a large amount of stripe cannot be removed after treatment with the comparison methods using SLD, WFAF, and LRTD. Furthermore, a small amount of stripe residue remains after applying the UTV method, and there are inevitable distortions. The MsDUC destriping results seem relatively good; however, the method has difficulty handling the different types of stripes in Case 3 and retains some residual striping in Case 4; thus, the overall results are not satisfactory. The processing effects of the SGIDN method are better in Cases 3 and 4, but this approach is slightly inferior to the proposed method in terms of spectral

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TABLE I

QUANTITATIVE STRIPE REDUCTION RESULTS OBTAINED FROM THE SIMULATED EXPERIMENTS

	Stripe Level	Index	SLD	UTV	WFAF	LTDR	MsDUC	SGIDN	HSI-DeNet	En-OCTNet
Case 1	20	MPSNR	34.37	29.52	39.07	39.30	43.85	43.53	45.02	47.86
		MSSIM	0.6884	0.6823	0.9096	0.9326	0.9466	0.9446	0.9579	0.9884
		MSAM	3.01	3.65	3.27	3.13	3.83	4.64	5.91	2.02
	30	MPSNR	35.52	29.19	36.84	37.99	40.48	41.29	43.13	45.87
		MSSIM	0.6832	0.5837	0.8126	0.9073	0.9364	0.9279	0.9555	0.9716
		MSAM	4.14	5.94	3.64	3.02	4.16	4.98	5.22	1.95
	50	MPSNR	33.97	26.32	30.58	35.56	37.68	37.66	38.79	40.93
		MSSIM	0.6380	0.5643	0.6161	0.8940	0.9116	0.9124	0.9261	0.9363
		MSAM	5.01	4.86	14.78	3.11	7.75	5.17	5.52	2.48
Case 2	(0, 50)	MPSNR	30.74	29.36	32.89	32.91	34.63	33.27	34.97	36.85
		MSSIM	0.7613	0.6453	0.7701	0.8739	0.8711	0.8531	0.9026	0.9097
		MSAM	6.89	6.64	5.36	3.67	4.70	4.84	4.58	3.54
Case 3	(0, 20)	MPSNR	33.47	27.94	35.50	34.89	36.84	33.18	38.61	40.95
		MSSIM	0.8568	0.8764	0.8475	0.8804	0.9081	0.8530	0.9230	0.9841
		MSAM	4.06	3.33	4.29	3.21	3.39	4.83	4.46	3.03
Case 4	(0, 20)	MPSNR	34.10	30.74	36.69	31.42	35.41	33.28	34.10	39.91
		MSSIM	0.9087	0.7871	0.8901	0.7336	0.8887	0.8479	0.8780	0.9093
		MSAM	3.59	4.12	3.72	3.27	4.27	4.92	4.77	3.64
Case 5	R_ (0, 20) S_ (0, 20)	MPSNR	22.89	19.37	21.61	27.98	31.86	29.51	30.99	33.71
		MSSIM	0.2950	0.2179	0.2882	0.6208	0.8987	0.8521	0.8914	0.9019
		MSAM	14.31	13.25	14.07	13.77	3.23	2.48	4.49	2.16

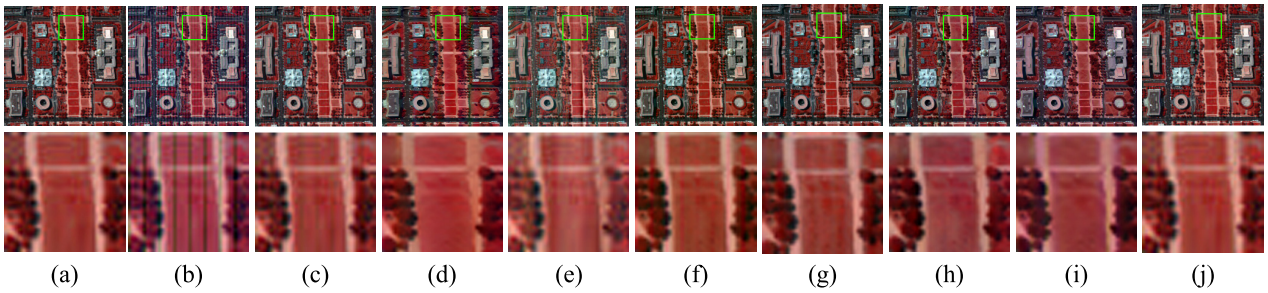


Fig. 4. Simulated destriped results for Case 1, with $\sigma = 20$. The picture in the second row is an enlargement of the green rectangle in the first row. (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

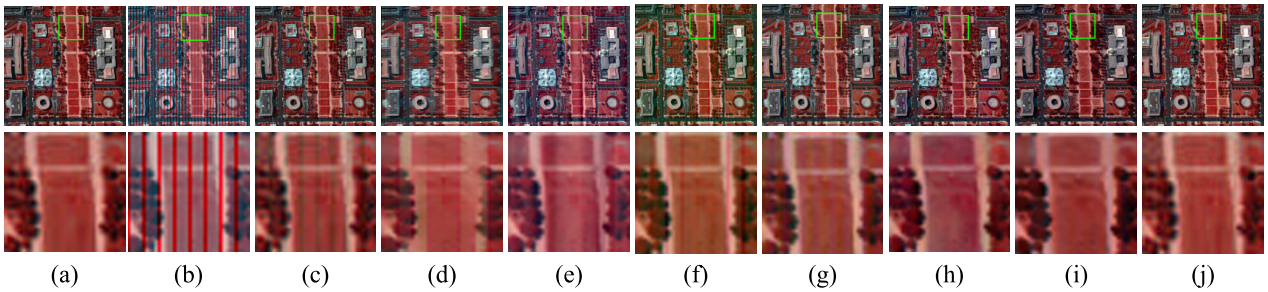


Fig. 5. Simulated destriped results for Case 1, with $\sigma = 50$. The picture in the second row is an enlargement of the green rectangle in the first row. (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

information restoration. It is obvious that, after the HSI-DeNet processing, most of the stripes can be basically removed from the visual effect, yet subtle stripes remain that cannot be fully removed. At the same time, the spectral information of the repaired image is also lost to a certain extent. Stripes mixed with random noise increase the difficulty of processing due to the discrete distribution of random noise, which is one of the difficulties of HSI data destriping. As shown in Fig. 9, the comparison method is capable of destriping to a certain extent, but the removal of random noise is not satisfactory. In the comparison method, although the destriping effect of HSI-DeNet is better, part of the image detail information of the ground features is oversmoothed and lost. However, the proposed method removes both stripe and random noise.

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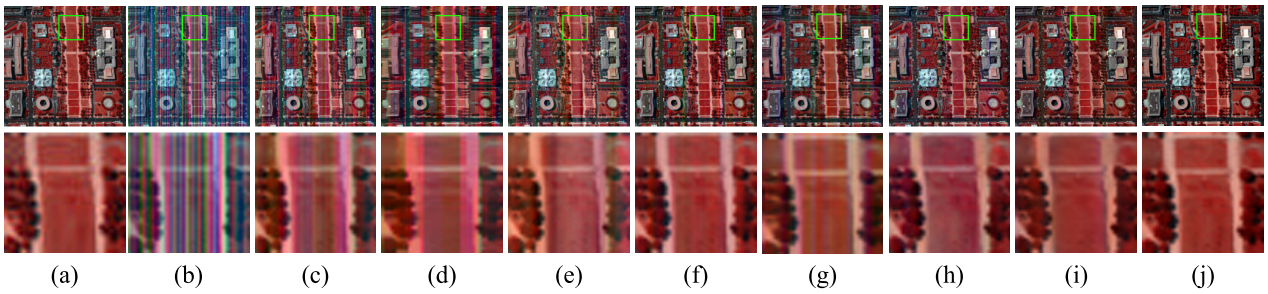


Fig. 6. Simulated destriped results for Case 2. (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

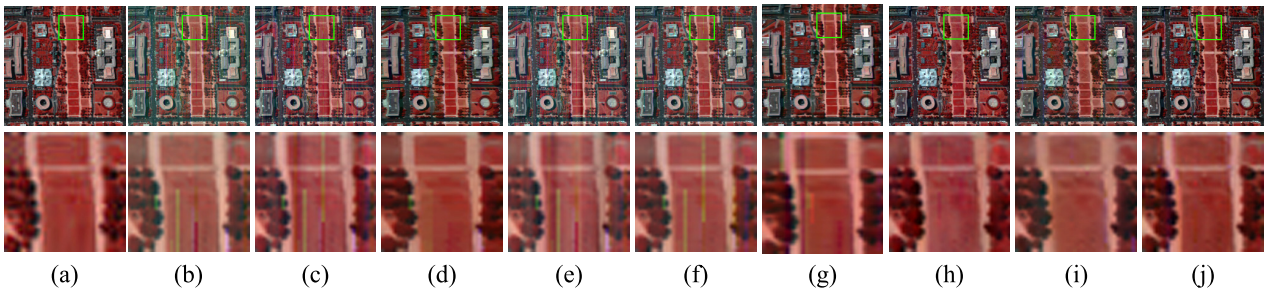


Fig. 7. Simulated destriped results for Case 3. (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

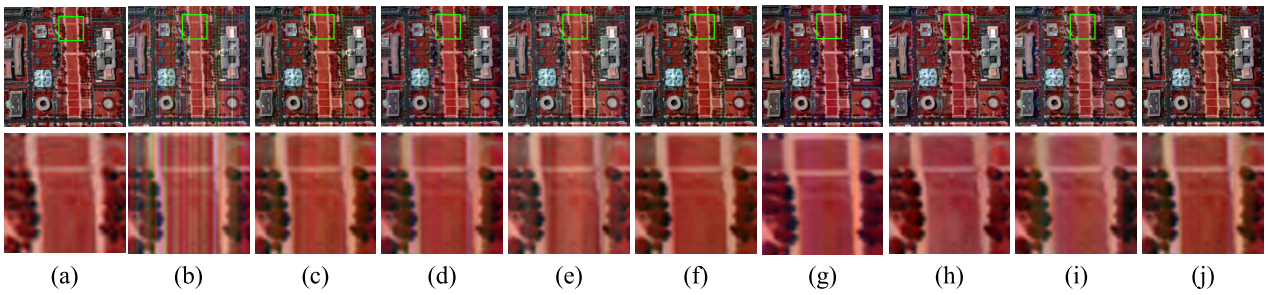


Fig. 8. Simulated destriped results for Case 4. (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

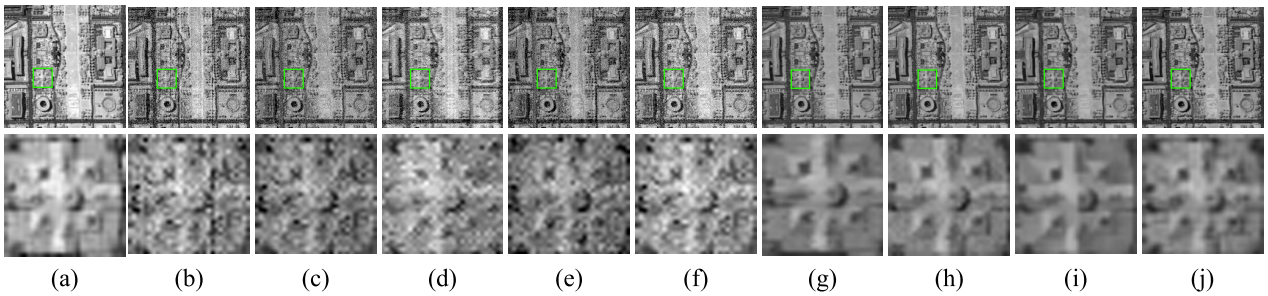
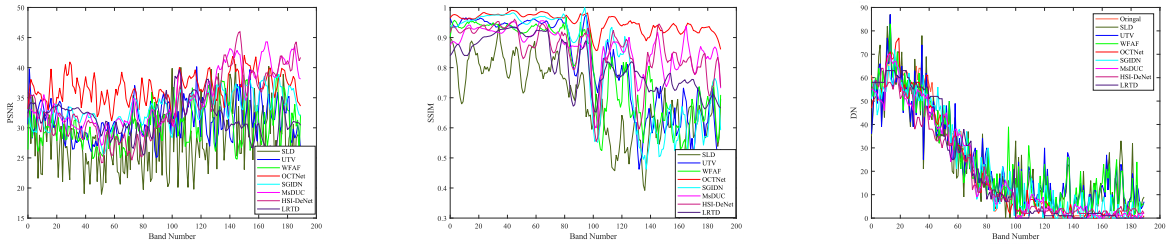


Fig. 9. Simulated destriped results for Case 5 (band 62). (a) Original image. (b) Degraded image. (c) SLD. (d) UTV. (e) WFAF. (f) LRTD. (g) MsDUC. (h) SGIDN. (i) HSI-DeNet. (j) En-OCTNet.

Fig. 10 presents the band-by-band PSNR and SSIM, as well as the gray value of the restored image in Case 2. All three of these quantities demonstrate that the proposed method could obtain excellent results in each band and more intuitively illustrate the advantages of the proposed destriping method. Furthermore, the results are best among the considered methods in terms of spectral information retention and restoration. In summary, En-OCTNet demonstrated significant advantages in various types of simulation experiments because it considers global context information. The network model can fully utilize long-distance dependencies under the guidance of attention. Such key information enables the model

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(a) (b) (c)

Fig. 10. Parameters PSNR, SSIM, and DN of different bands obtained using considered methods in Case 2. (a) PSNR of each band. (b) SSIM of each band. (c) DN of the pixel at (16, 37).

TABLE II

QUANTITATIVE STRIPE REDUCTION RESULTS OBTAINED IN THE OHS-01 REAL DATA EXPERIMENTS

Methods	Scene 1		Scene 2		Scene 3	
	ICV	MRD (%)	ICV	MRD (%)	ICV	MRD (%)
SLD	11.80	1.43	4.53	4.68	2.84	0.49
UTV	7.88	4.30	4.82	5.82	2.91	0.23
WAFA	9.77	3.28	4.54	4.35	2.88	0.24
LRTD	10.46	1.94	4.86	1.19	3.76	0.51
MsDUC	11.54	1.91	4.26	1.50	3.54	0.72
SGIDN	6.19	5.34	4.37	11.32	2.79	0.61
HSI-DeNet	11.39	1.59	4.28	2.01	3.47	0.94
En-OCTNet	13.27	1.55	4.97	1.37	3.04	0.99

to identify not only "what to look for" but also "where to look at" in order to learn the key feature information that distinguishes stripe- and stripe-free image components.

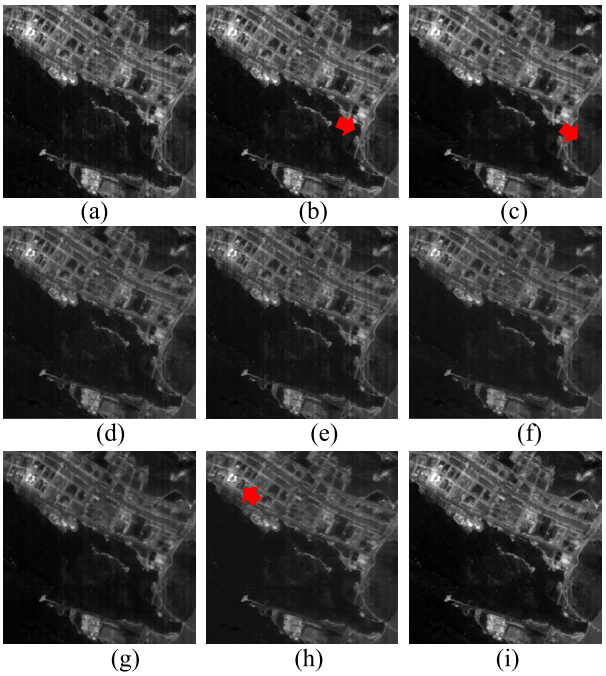
D. Results of Real Data Experiments

To verify the applicability of the proposed method to real data, we conducted experiments on four sets of real data. We selected three scenes of OHS-01, which were obtained with 32 bands, all of size 512 × 512, and one Earth Observing-1 (EO-1) image scene (https://pan.baidu.com/s/14ht8k5H-8ObzHJS6msYZjQ), obtained using the EO-1 Hyperion sensor, with a size of 200 × 200 × 198.

1) OHS-01: Scenes 1 and 2 correspond to band 22, and Scene 3 corresponds to band 30. The features covered in the three scenes include water, mountains, buildings, and clouds. The stripes in the images show obvious spatial distribution differences. The stripes in Scenes 1 and 2 are almost invisible on the mountain, showing low brightness, and are hidden in the background, but they exhibit high brightness on the buildings and water bodies. In contrast, the stripes in Scene 3 are visible on the clouds, whereas, on the water body, the stripes blend in with and are indistinguishable from the background.

Therefore, the OHS-01 data stripes are more complicated and representative. For this experiment, we used the evaluation indicators ICV and MRD, and Table II presents the evaluation index results. Figs. 11–13 depict the results for each scene, respectively.

For Scene 1, as shown in Fig. 11, the SLD method does not evidently suppress the stripes overall. Processing by UTV removes the stripes in the homogeneous area to a certain extent but has almost no effect on the area with a complex texture



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 11. Real destriped results for OHS-01 data of Scene 1. (a) Original image. (b) SLD. (c) UTV. (d) WFAF. (e) LRTD. (f) MsDUC. (g) SGIDN. (h) HSI-DeNet. (i) En-OCTNet.

structure, as shown by the arrow in Fig. 11(c). Processing by WFAF and LRTD suppresses most of the stripes, but stripes remain to some extent in the details. Meanwhile, the MsDUC, SGIDN, and HSI-DeNet methods suppress the stripes to a certain extent but are not sufficiently thorough, while the restored images are excessively smooth due to a certain amount of spectral loss. In contrast, En-OCTNet achieves the best destriping performance, can effectively remove stripes, and simultaneously retains the image spectrum and details.

For Scene 2, as shown in Fig. 12, there is a large amount of dense stripe interference. The SLD, UTV, and WFAF methods do not completely suppress the stripes. In the WAFA and SLD results, there are obvious stripes in the mountainous area. Simultaneously, the UTV method loses detailed information about the image structure. The result of LRTD seems relatively good, but some stripes remain. The MsDUC results are not satisfactory for Scene 2, where the remaining stripes are obvious. The HSI-DeNet method can suppress the overwhelming majority of stripes, but it does not completely suppress all

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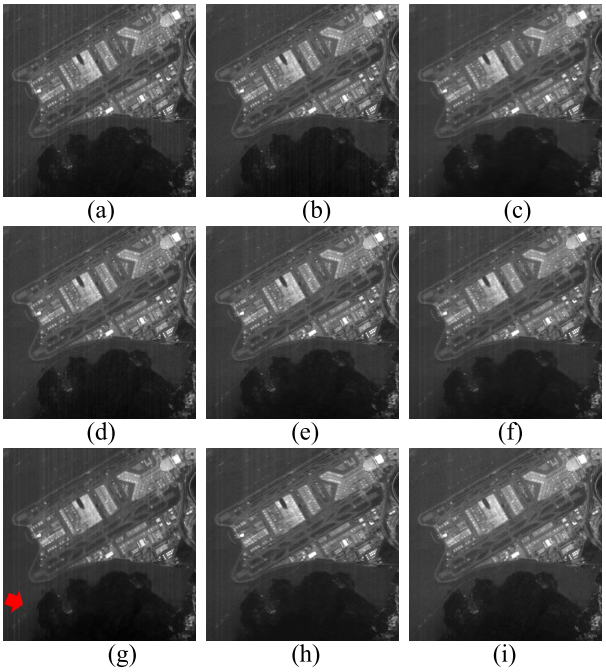


Fig. 12. Real destriped results for OHS-01 data of Scene 2. (a) Original image. (b) SLD. (c) UTV. (d) WFAF. (e) LRTD. (f) MsDUC. (g) SGIDN. (h) HSI-DeNet. (i) En-OCTNet.

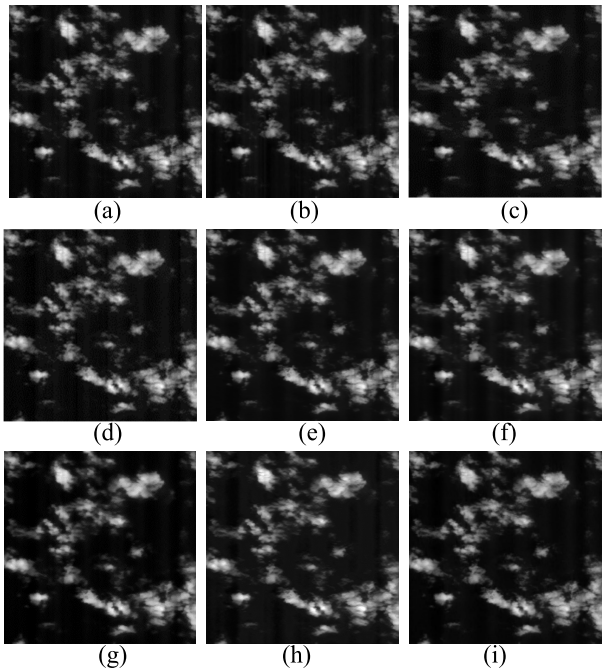


Fig. 13. Real destriped results for OHS-01 data of Scene 3. (a) Original image. (b) SLD. (c) UTV. (d) WFAF. (e) LRTD. (f) MsDUC. (g) SGIDN. (h) HSI-DeNet. (i) En-OCTNet.

stripes. In contrast, the proposed method has significant advantages in terms of evaluation indicators: the stripe suppression is obvious, the details and spectral information of ground objects are preserved completely, and the destriping result is best among all the methods.

For Scene 3, as shown in Fig. 13, the stripes blend in with the black water background, making it difficult to accurately distinguish between the stripes and water background and,

TABLE III

QUANTITATIVE STRIPE REDUCTION RESULTS OBTAINED FROM THE EO-01 REAL DATA EXPERIMENTS

Methods	EO-01	
	MICV	MMRD (%)
SLD	7.74	9.74
UTV	8.18	9.01
WAFA	8.25	9.73
LRTD	8.32	8.62
MsDUC	8.46	9.52
SGIDN	9.75	9.13
HSI-DeNet	8.85	7.63
En-OCTNet	9.99	8.43

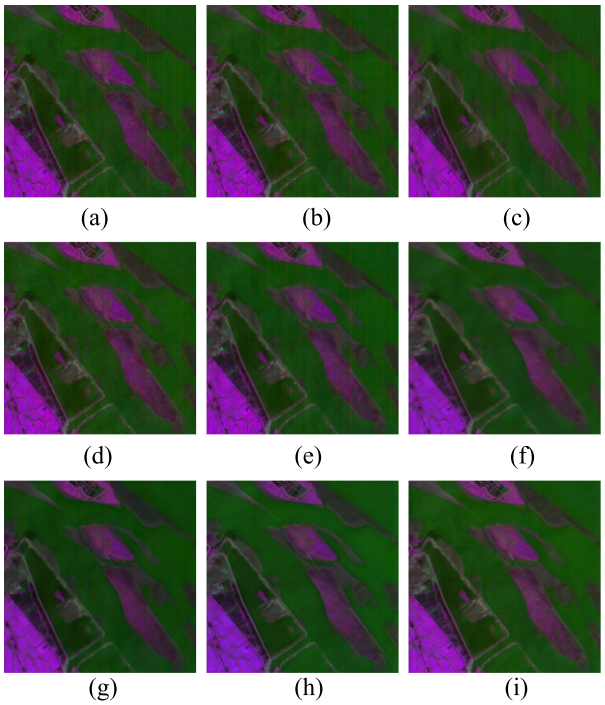


Fig. 14. Real destriped results obtained using EO-01 data. (a) Original image. (b) SLD. (c) UTV. (d) WFAF. (e) LRTD. (f) MsDUC. (g) SGIDN. (h) HSI-DeNet. (i) En-OCTNet.

consequently, increasing the difficulty of destriping. According to the experimental results, the seven contrast methods do not completely remove the stripes. The clouds retain obvious stripes, and there is also a certain amount of loss of radiation information. The proposed method can mine the key features of the stripe and stripe-free components, and remove stripes accurately while retaining the original information. In contrast, the results of the three traditional destriping methods are not ideal, nor are they suitable for complex stripe types. The two deep learning methods do not adequately consider both the effects of destriping and the preservation of ground object details.

2) EO-01: A river channel image with a size of 200 × 200 × 198 that was contaminated by stripe and random noise was used in this experiment, and the proposed method was used for stripe processing. Fig. 14 displays the experimental results in the form of pseudocolor images (bands 72, 22, and 42), and Table III lists the index evaluation results. As shown in Fig. 14, the proposed method yields

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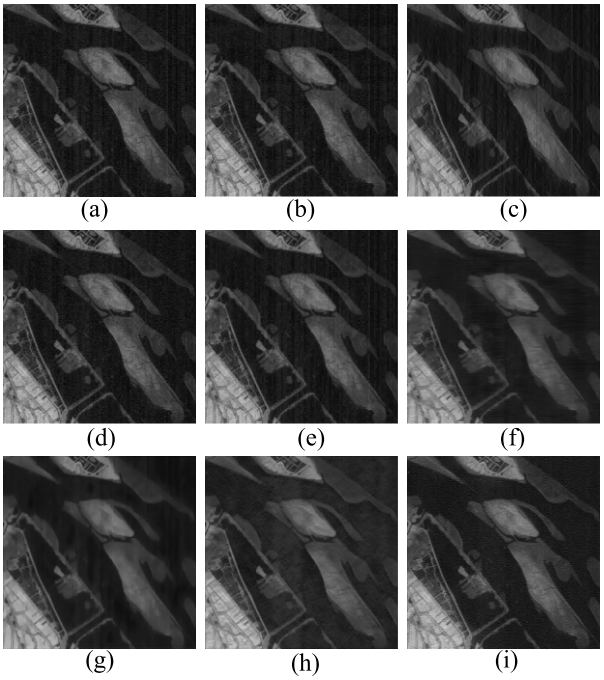


Fig. 15. EO-01 data destriped result (band 56). (a) Original image. (b) SLD. (c) UTV. (d) WFAF. (e) LRTD. (f) MsDUC. (g) SGIDN. (h) HSI-DeNet. (i) En-OCTNet.

reasonable destriping results, especially in terms of image detail and spectral information restoration. However, stripes remain after destriping using the SLD, UTV, WFAF, and LRTD approaches. Of these, the WFAF technique suppressed some of the stripes, but some spectral information was lost in the homogeneous area, as shown in Fig. 14(d). After applying the MsDUC method, the stripes were generally suppressed, but a loss of spectral information occurred; thus, the destriped image is oversmoothed. The SGIDN method caused a loss of structural information in the repaired image, also resulting in oversmoothness. The HSI-DeNet method suppressed the stripes and restored the detail information but changed the original spectral information.

To compare the effects of different methods more intuitively, we selected band 56 to show the destriping results, as depicted in Fig. 15. Band 56 is more severely affected by stripe noise, and the stripes are densely distributed and have random widths, making this band a representative example.

It can be seen from the results that the visual effect of SLD and LRTD processing is not good: only a small amount of stripe noise is slightly suppressed, and many obvious stripes remain. After UTV processing, some of the stripe interference can be relieved, but the texture information of the image structure is lost, and the restored image is excessively smoothed. The WAFA method generally removes the stripes, but a few stripes are not removed locally. In addition, the MsDUC, SGIDN, and HSI-DeNet restoration results possess obvious stripes that were not removed, and for all methods, the destriping caused a loss of image structure information. The En-OCTNet method yields the best destriping results among the six methods, and the restored image retains the original structure and spectral information of the ground features.

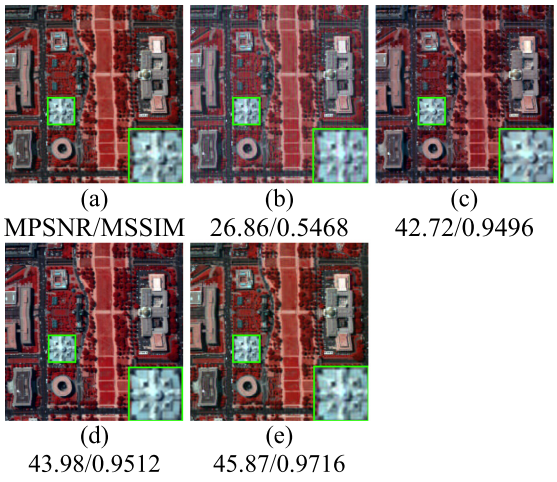


Fig. 16. Destriping results for different convolutions. (a) Original image. (b) Degraded image. (c) V-ConvNet. (d) En-OCTNet-P. (e) En-OCTNet.

In short, destriping using the proposed method produced a satisfactory image result.

E. Discussion

In this section, we consider Case 1 with $\sigma = 30$ as an example to demonstrate the effectiveness and superiority of the cross-frequency feature extraction module and CESS feature fusion module proposed in this article. In addition, the influence of the number of bands of the input data on the destriping effect and the operating efficiency of the proposed method are also discussed. Moreover, to illustrate the generalizability of the proposed method, we performed experiments on two infrared images with stripe noise and one natural image with random noise. The details of the following discussion strongly prove the advantages of the proposed method, as follows.

1) Effectiveness of the Cross-Frequency Feature Extraction Module: Here, we focus on discussing the destriping performance in the network structure of this article, using traditional convolution (denoted as V-ConvNet), the original Oct-Conv with pooling operation (denotes as En-OCTNet-P), and the En-OCTNet as a feature extractor. The experimental results are shown in Fig. 16.

From Fig. 16, the traditional convolution has a certain relief effect on the stripe, but the destriping effect overall is not good, as the stripe cannot be completely removed. At the same time, the destriped image spectral information is different from the clean image, described as follows. The En-Oct-Conv-P extracts and retains low-frequency information by pooling and cannot achieve satisfactory destriping results, as the detailed spectral information is lost and experiences the oversmoothing phenomenon in the image, as shown in Fig. 16(d). In contrast, En-OCTNet has significant advantages. As shown in Fig. 16(e), at the same stripe level, most stripes are removed, and better restored images can be obtained. Therefore, the proposed method has obvious advantages in stripe removal and spectral information retention.

2) Effectiveness of the CESS: This section discusses the contribution of the CESS feature fusion module to stripe removal. In particular, we also compared the CESS feature

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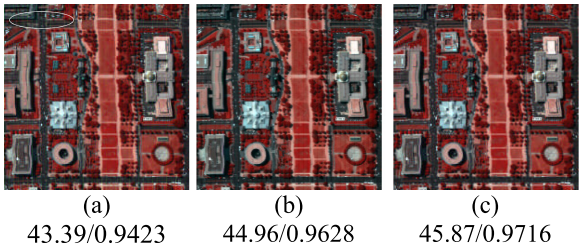


Fig. 17. Destriping results for different attentions. (a) OCTNet-N. (b) OCTNet-dual. (c) En-OCTNet.

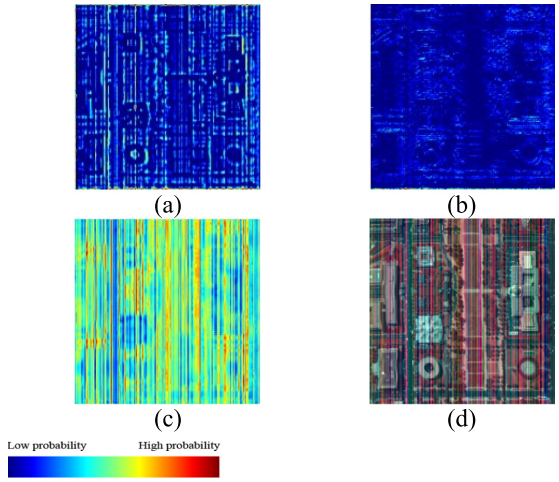


Fig. 18. Output feature of CESS module. (a) High-frequency feature map in layer 4. (b) High-frequency feature map in layer 4. (c) Visualization of the attention feature map. The color depth indicates the probability of being stripe.

fusion module with spatial–spectral attention commonly used in HSI processing. Here, the OCTNet-N denotes the proposed method without the CESS feature fusion module, and the OCTNet-dual denotes the CESS feature fusion module replaced by spatial–spectral attention [57]. The experimental results are shown in Fig. 17. Moreover, we also display the output feature map of the CESS module, as shown in Fig. 18.

As shown in Fig. 17, these three methods have a certain destriping effect. However, as shown in Fig. 17(a), some subtle stripes remain, especially at the upper and lower edges of the image, as shown in the ellipse-marked area. These residual stripes are due to the fact that, although Dila-Conv can expand the receptive field to a certain extent and obtain contextual information within the range of the receptive field, it cannot mine deeper into the key features that distinguish stripe and nonstripe information from a perspective of the global receptive field. OCTNet-dual, as shown in Fig. 17(b), shows good stripe removal performance; however, the performances of MPSNR and MSSIM are lower than that of En-OCTNet. In short, the proposed method can retain the original image information better and capture the global context information through Dila-Conv and global context attention, such that the network knows more precisely "what to look for" and "where to look at" and achieves a balance between destriping and preservation of the stripe-free image components.

As shown in Fig. 18(b), the low-frequency branch feature map completely maintained the main information of the image.

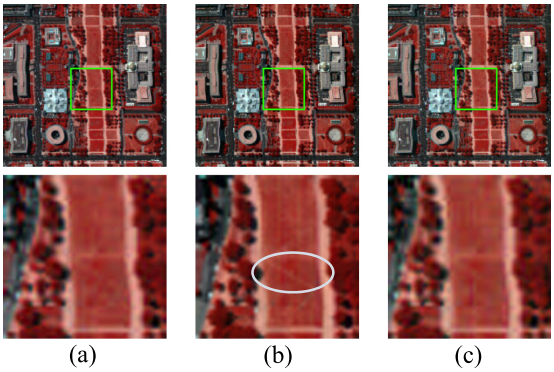


Fig. 19. Destriping results for different input bands. (a) Original image. (b) En-OCTNet-S. (c) En-OCTNet.

After the CESS module, as shown in Fig. 18(c), most of the stripe patterns are detected, indicating that the attention layer is able to provide supportive information to the model, which can guide the model to focus on learning the features to discriminate stripe and stripe-free components.

3) Effectiveness of the Input Band: This section discusses the impact of the number of input bands of the training data on stripe removal. We used single-band input (denoted as En-OCTNet-S) and ten-band input for network training, respectively. The experimental results are shown in Fig. 19.

From Fig. 19, En-OCTNet-S has an obviously strong inhibitory effect on the stripe. The overall tone is not much different from that of the original image, but it can be seen from the enlarged image that En-OCTNet-S is slightly inferior to En-OCTNet in terms of spectral information retention and detail restoration; furthermore, a small amount of stripe residue can be identified. Therefore, the use of multiband input instead of single-band input is necessary and advantageous for the restoration of the degraded image.

4) Time Costs: This section compares the destriping time consumptions of different methods. All experiments were performed on a PC with Windows 10, 32-GB RAM, and an NVIDIA GeForce gtx2080 Ti GPU. Here, EO-01 from the real data experiment is taken as an example for the analysis. In addition, we know that the trainable parameters of the model have a critical impact on the inference speed of the model. Therefore, we have produced statistics on the number of trainable parameters of deep learning methods. Table IV shows the time costs and the number of trainable parameters of the various methods. Recall that En-OCTNet utilizes a two-branch structure to extract and separate cross-frequency features that can reduce model parameters. Moreover, in the forward propagation stage, fewer calculation parameters are required. Therefore, under the same experimental conditions, the proposed destriping method achieves the least time consumption.

Trainable parameters and giga-floating-point operations per second systems (GFLOPs) are theoretically inversely proportional to model inference time. Unfortunately, model inference time is affected by many factors, such as the operating platform, memory access cost, and deep learning framework. Therefore, the trainable parameters and GFLOPs cannot be

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TABLE IV
TIME COSTS OF DIFFERENT DESTRIPING METHODS IN EO-01 DATA EXPERIMENTS

Method	Time (s)	Parameters (M)	GFLOPs
SLD	8.6	-	-
UTV	10.1	-	-
WAFA	9.4	-	-
LRTD	7.3	-	-
MsDUC	2.5	0.17	6.94
SGIDN	9.6	1.73	50.42
HSI-DeNet	4.8	2.75	89.82
En-OCTNet	3.9	0.71	28.36

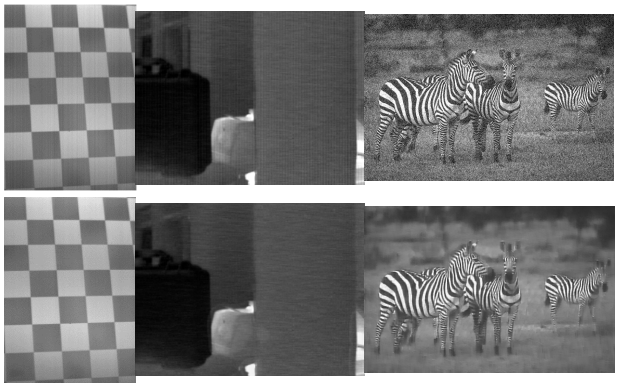


Fig. 20. Experimental generalization of the proposed method. (a) and (b) Infrared images. (c) Natural image.

used as direct indicators to measure the speed of the model but only as reference indicators.

5) Generalization Study: To illustrate the generalizability of the proposed method, we performed experiments on two groups of infrared images with stripe noise and a natural image with random noise. The two infrared images were acquired from the online source https://github.com/zhonglingyuxiuYYX/DLNUC/tree/master/Data, while the natural image was obtained from https://github.com/wbhu/DnCNN-tensorflow. Fig. 20 presents the experimental results, demonstrating that the proposed method could remove stripe and random noise from the images to achieve satisfactory visual results while, at the same time, retaining the structural and textural details of the images. It is worth noting that the imaging technology used to acquire the images and the cause of stripes in the different images are different. Moreover, the model was developed by training hyperspectral remote sensing image data. Hence, the infrared and natural images are unknown datasets for the destriping model, which poses a considerable challenge. However, the model still achieves satisfactory destriping without performing other operations, such as fine-tuning, which strongly illustrates the versatility of the proposed method. The results of this generalization study are mainly possible because the model separates the stripe- and stripe-free components through convolution; it learns both the stripe characteristics and stripe-free features. Simultaneously, the proposed model understands and learns the key distinguishing features under the guidance of the attention mechanism such that the model focuses on more learning, that is, the model knows not only "what to look for" but also "where to look at."

V. CONCLUSION

This article proposes En-OCTNet, an HSI data stripe removal method utilizing a CNN based on Oct-Conv that can extract the cross-frequency features of a feature map. In this method, Dila-Conv and the cross channel enhanced spatial–spectral feature fusion module are applied to learn the discriminative properties between the stripe and stripe-free components, thereby providing a new means of HSI data stripe removal. Experiments were conducted using a simulation with multiple stripe types and real HSI data corresponding to two different satellite images. Moreover, the results obtained by applying the proposed approach were compared with those acquired using the existing HSI data destriping and denoising methods. The comparative results indicate that the proposed method can handle different types of stripes and obtain state-of-the-art results. We also analyzed the contributions and roles of the key modules in the proposed method to demonstrate its efficiency and generalizability. The ablation experiment results show that, benefited by the cross-frequency feature extraction of Oct-Conv, the spectral information of stripe-free components in HSI can be well preserved, and the information loss of the destriped image can be reduced. In summary, the experimental results reveal the advantages of the proposed method in stripe removal and image restoration. In future work, we will further investigate the application of multiscale characteristics in HSI data destriping and explore blind HSI data destriping combined with real data and hybrid noise removal. Through deep learning, we will attempt to obtain restoration images directly from the real data contaminated by stripe noise.

REFERENCES

[1] O. Sidorov and J. Y. Hardeberg, "Deep hyperspectral prior: Single-image denoising, inpainting, super-resolution," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3844–3851.
[2] Q. Zhang, Q. Yuan, J. Li, F. Sun, and L. Zhang, "Deep spatio–spectral Bayesian posterior for hyperspectral image non-i.i.d. noise removal," *ISPRS J. Photogramm. Remote Sens.*, vol. 164, pp. 125–137, Jun. 2020.
[3] H. Cui *et al.*, "Multiscale intensity propagation to remove multiplicative stripe noise from remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2308–2323, Apr. 2020.
[4] Y. Chang, L. Yan, H. Fang, and C. Luo, "Anisotropic spectral–spatial total variation model for multispectral remote sensing image destriping," *IEEE Trans. Image Process.*, vol. 24, no. 6, pp. 1852–1866, Jun. 2015.
[5] R. Chen, W. Yu, R. Wang, G. Liu, and Y. Shao, "Interferometric phase denoising by pyramid nonlocal means filter," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 4, pp. 826–830, Jul. 2013.
[6] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, "Image denoising by sparse 3-D transform-domain collaborative filtering," *IEEE Trans. Image Process.*, vol. 16, no. 8, pp. 2080–2095, Aug. 2007.
[7] J. G. Liu and G. L. K. Morgan, "FFT selective and adaptive filtering for removal of systematic noise in ETM+ imageodesy images," *IEEE Trans. Geosci. Remote Sens.*, vol. 44, no. 12, pp. 3716–3724, Dec. 2006.
[8] A. Duijster, P. Scheunders, and S. De Backer, "Wavelet-based EM algorithm for multispectral-image restoration," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 11, pp. 3892–3898, Nov. 2009.

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[9] H. Carfantan and J. Idier, "Statistical linear destriping of satellite-based pushbroom-type images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 4, pp. 1860–1871, Apr. 2010.

[10] H. Zhang, L. Liu, W. He, and L. Zhang, "Hyperspectral image denoising with total variation regularization and nonlocal low-rank tensor decomposition," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3071–3084, May 2020.

[11] W. Xie and Y. Li, "Hyperspectral imagery denoising by deep learning with trainable nonlinearity function," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 11, pp. 1963–1967, Nov. 2017.

[12] Y. Chen *et al.*, "Drop an octave: Reducing spatial redundancy in convolutional neural networks with octave convolution," 2019, *arXiv:1904.05049*.

[13] B. K. P. Horn and R. J. Woodham, "Destriping LANDSAT MSS images by histogram modification," *Comput. Graph. Image Process.*, vol. 10, no. 1, pp. 69–83, May 1979.

[14] G. Corsini, M. Diani, and T. Walzel, "Striping removal in MOS-B data," *IEEE Trans. Geosci. Remote Sens.*, vol. 38, no. 3, pp. 1439–1446, May 2000.

[15] F. L. Gadallah, F. Csillag, and E. J. M. Smith, "Destriping multisensor imagery with moment matching," *Int. J. Remote Sens.*, vol. 21, no. 12, pp. 2505–2511, Jan. 2000.

[16] M. Zhang and B. K. Gunturk, "Multiresolution bilateral filtering for image denoising," *IEEE Trans. Image Process.*, vol. 17, no. 12, pp. 2324–2333, Dec. 2008.

[17] P. Rakwatin, W. Takeuchi, and Y. Yasuoka, "Stripe noise reduction in MODIS data by combining histogram matching with facet filter," *IEEE Trans. Geosci. Remote Sens.*, vol. 45, no. 6, pp. 1844–1856, Jun. 2007.

[18] Z.-W. Li, X.-L. Ding, D.-W. Zheng, and C. Huang, "Least squares-based filter for remote sensing image noise reduction," *IEEE Trans. Geosci. Remote Sens.*, vol. 46, no. 7, pp. 2044–2049, Jul. 2008.

[19] H. Yu, L. Zhao, and H. Wang, "Image denoising using trivariate shrinkage filter in the wavelet domain and joint bilateral filter in the spatial domain," *IEEE Trans. Image Process.*, vol. 18, no. 10, pp. 2364–2369, Oct. 2009.

[20] N. Liu, W. Li, R. Tao, and J. E. Fowler, "Wavelet-domain low-rank/group-sparse destriping for hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 12, pp. 10310–10321, Dec. 2019.

[21] J. Xue, Y. Zhao, W. Liao, and S. G. Kong, "Joint spatial and spectral low-rank regularization for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1940–1958, Apr. 2018.

[22] T. Xie, S. Li, and B. Sun, "Hyperspectral images denoising via nonconvex regularized low-rank and sparse matrix decomposition," *IEEE Trans. Image Process.*, vol. 29, pp. 44–56, 2020.

[23] Q. Yuan, L. Zhang, and H. Shen, "Hyperspectral image denoising employing a spectral–spatial adaptive total variation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 10, pp. 3660–3677, Oct. 2012.

[24] M. Protter and M. Elad, "Image sequence denoising via sparse and redundant representations," *IEEE Trans. Image Process.*, vol. 18, no. 1, pp. 27–35, Jan. 2009.

[25] Y.-Q. Zhao and J. Yang, "Hyperspectral image denoising via sparse representation and low-rank constraint," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 1, pp. 296–308, Jan. 2015.

[26] W. He, H. Zhang, H. Shen, and L. Zhang, "Hyperspectral image denoising using local low-rank matrix recovery and global spatial–spectral total variation," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 3, pp. 713–729, Mar. 2018.

[27] M. Wang, Q. Wang, J. Chanussot, and D. Li, "Hyperspectral image mixed noise removal based on multidirectional low-rank modeling and spatial–spectral total variation," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 488–507, Jan. 2021.

[28] Y. Tao, M. Xu, F. Zhang, B. Du, and L. Zhang, "Unsupervised-restricted deconvolutional neural network for very high resolution remote-sensing image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 12, pp. 6805–6823, Dec. 2017.

[29] G. Deng, Z. Wu, C. Wang, M. Xu, and Y. Zhong, "CCANet: Class-constraint coarse-to-fine attentional deep network for subdecimeter aerial image semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–20, 2022, doi: 10.1109/TGRS.2021.3055950.

[30] Y. Gong *et al.*, "Context-aware convolutional neural network for object detection in VHR remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 34–44, Jan. 2020.

[31] Y. Chang, M. Chen, L. Yan, X.-L. Zhao, Y. Li, and S. Zhong, "Toward universal stripe removal via wavelet-based deep convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2880–2897, Apr. 2020.

[32] W. Liu and J. Lee, "A 3-D atrous convolution neural network for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 8, pp. 5701–5715, Aug. 2019.

[33] Q. Yuan, Q. Zhang, J. Li, H. Shen, and L. Zhang, "Hyperspectral image denoising employing a spatial–spectral deep residual convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 1205–1218, Feb. 2019.

[34] A. Maffei, J. M. Haut, M. E. Paoletti, J. Plaza, L. Bruzzone, and A. Plaza, "A single model CNN for hyperspectral image denoising," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 4, pp. 2516–2529, Apr. 2020.

[35] K. Zhang, W. Zuo, and L. Zhang, "FFDNet: Toward a fast and flexible solution for CNN-based image denoising," *IEEE Trans. Image Process.*, vol. 27, no. 9, pp. 4608–4622, Sep. 2018.

[36] Y. Chang, L. Yan, H. Fang, S. Zhong, and W. Liao, "HSI-DeNet: Hyperspectral image restoration via convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 2, pp. 667–682, Feb. 2019.

[37] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, "Beyond a Gaussian denoiser: Residual learning of deep CNN for image denoising," *IEEE Trans. Image Process.*, vol. 26, no. 7, pp. 3142–3155, Jul. 2017.

[38] Y. Zhong, W. Li, X. Wang, S. Jin, and L. Zhang, "Satellite-ground integrated destriping network: A new perspective for EO-1 Hyperion and Chinese hyperspectral satellite datasets," *Remote Sens. Environ.*, vol. 237, Feb. 2020, Art. no. 111416.

[39] Q. Zhang, Q. Yuan, J. Li, X. Liu, H. Shen, and L. Zhang, "Hybrid noise removal in hyperspectral imagery with a spatial–spectral gradient network," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 10, pp. 7317–7329, Oct. 2019.

[40] Q. Shi, X. Tang, T. Yang, R. Liu, and L. Zhang, "Hyperspectral image denoising using a 3-D attention denoising network," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 12, pp. 10348–10363, Dec. 2021, doi: 10.1109/TGRS.2020.3045273.

[41] H. Ma, G. Liu, and Y. Yuan, "Enhanced non-local cascading network with attention mechanism for hyperspectral image denoising," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2448–2452, doi: 10.1109/ICASSP40776.2020.9054630.

[42] Y. Zhao, D. Zhai, J. Jiang, and X. Liu, "ADRN: Attention-based deep residual network for hyperspectral image denoising," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 2668–2672, doi: 10.1109/ICASSP40776.2020.9054658.

[43] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2018.

[44] J. X. Y. Cao, S. Lin, F. Wei, and H. Hu, "GCNet: Non-local networks meet squeeze-excitation networks and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1971–1980.

[45] X. Wang, R. Girshick, A. Gupta, and K. He, "Non-local neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2018, pp. 7794–7803.

[46] J. Hu, L. Shen, and G. Sun, "Squeeze-and-excitation networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jul. 2018, pp. 7132–7141.

[47] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. ICLR*, 2015, pp. 1–15.

[48] A. Paszke *et al.*, "Automatic differentiation in PyTorch," in *Proc. 31st Conf. Neural Inf. Process. Syst.*, Long Beach, CA, USA, 2017, pp. 1–4.

[49] R. Pande-Chhetri and A. Abd-Elrahman, "De-striping hyperspectral imagery using wavelet transform and adaptive frequency domain filtering," *ISPRS J. Photogramm. Remote Sens.*, vol. 66, no. 5, pp. 620–636, Sep. 2011.

[50] M. Bouali and S. Ladjal, "Toward optimal destriping of MODIS data using a unidirectional variational model," *IEEE Trans. Geosci. Remote Sens.*, vol. 49, no. 8, pp. 2924–2935, Aug. 2011.

[51] Y. Chen, T.-Z. Huang, and X.-L. Zhao, "Destriping of multispectral remote sensing image using low-rank tensor decomposition," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 11, no. 12, pp. 4950–4967, Dec. 2018.

[52] Z. Wang, G. Wang, Z. Pan, J. Zhang, and G. Zhai, "Fast stripe noise removal from hyperspectral image via multi-scale dilated unidirectional convolution," *Multimedia Tools Appl.*, vol. 79, nos. 31–32, pp. 23007–23022, Aug. 2020.

[53] Z. Wang, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

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[54] R. H. Yuhas, J. W. Boardman, and F. H. G. Alexander, "Determination of semi-arid landscape endmembers and seasonal trends using convex geometry spectral unmixing techniques," in *Proc. Summaries 4th Annu. JPL Airborne Geosci. Workshop*, vol. 1, Oct. 1993, pp. 205–208.
[55] G. M. Smith and P. J. Curran, "Methods for estimating image signal-to-noise ratio," in *Advances in Remote Sensing and GIS Analysis*, P. M. Atkinson and N. J. Tate, Eds. Hoboken, NJ, USA: Wiley, 2000, pp. 61–74.
[56] H. Shen and L. Zhang, "A MAP-based algorithm for destriping and inpainting of remotely sensed images," *IEEE Trans. Geosci. Remote Sens.*, vol. 47, no. 5, pp. 1492–1502, May 2009.
[57] J. Fu *et al.*, "Dual attention network for scene segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 3146–3154.



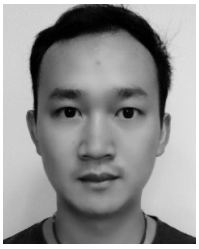
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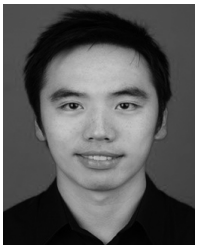
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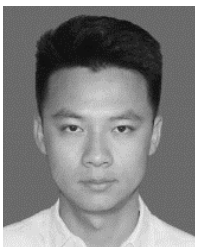
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